Project:
Mental Health Assisstant
(Depression Detection Using DistilBERT)

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1. Project Overview

This project presents a Mental Health Assistant designed to detect depression indicators in social media text. By fine-tuning the DistilBERT transformer model on a large multi-source dataset, the system classifies user-generated text as either depressive or non-depressive with high accuracy. The ultimate goal is to serve as an intelligent backbone for a conversational therapy assistant capable of identifying emotional distress in real time.

1.1 Objectives

- Build a robust binary classification model for depression detection from social media text.
- Perform thorough exploratory data analysis to uncover linguistic patterns associated with mental distress.
- Fine-tune DistilBERT on combined Reddit and Twitter datasets for optimal performance.
- Extend the system into a conversational AI therapy assistant using Ollama Llama 3.2 3B.

1.2 Workflow Summary

The project follows a structured ML pipeline:

(1) multi-source data ingestion

(2) data quality checks and cleaning

(3) exploratory data analysis

(4) text preprocessing and tokenization

(5) DistilBERT fine-tuning with early stopping

(6) evaluation on a held-out test set and an external dataset

(7) integration with a conversational LLM for deployment as a therapy assistant.

2. Data Summary and Description

Three publicly available datasets were sourced from Kaggle and combined to create a rich, diverse corpus for model training. The datasets span two major social media platforms Reddit and Twitter ensuring the model learns from varied writing styles, vocabularies, and expression patterns.

2.1 Source Datasets

Dataset	Source	Rows	Class Balance
Dataset 1	Reddit (infamouscoder) Kaggle link	7,731	Not Depression: 3,900 Depression: 3,831
Dataset 2	Twitter Train (samrats) Kaggle link	31,000	Not Depression: 15,000 Depression: 16,000
Dataset 3	Twitter Test (samrats) Kaggle link	31,000	Not Depression: 15,000 Depression: 16,000

2.2 Data Quality Overview

Dataset	Missing Values	Duplicate Rows
Reddit Dataset	None	81 rows
Twitter Train	None	932 rows
Twitter Test	None	1,956 rows

2.3 Combined Dataset

After concatenating the three datasets and standardising column names to 'text' and 'label', the combined dataset contained 69,731 rows. Following the removal of 2,969 duplicate records, the final cleaned dataset comprised 66,762 unique text samples with a near-balanced class distribution.

Metric	Value
Total Rows (raw)	69,731
Duplicate Rows Removed	2,969
Final Unique Rows	66,762
Not Depressed (Label 0)	~35,000 (~52%)
Depressed (Label 1)	~32,000 (~48%)

3. Data Cleaning Steps

A multi-step data cleaning pipeline was applied to ensure data quality and model readiness. Each step was deliberately designed to preserve emotional signal while removing noise.

3.1 Duplicate Removal

Exact duplicate rows were identified and removed. A total of 2,969 duplicate entries were found across the combined dataset. Removing duplicates prevents the model from over-fitting to repeated patterns and ensures each training sample contributes uniquely to learning.

3.2 Text Preprocessing for Modeling

A dedicated clean_text() function was applied to all text samples prior to training:

- URL Removal: All HTTP/HTTPS URLs were stripped using re.sub(r"http\S+", "", text) to eliminate irrelevant hyperlinks that carry no semantic value.
- HTML Tag Removal: Residual HTML markup was removed using re.sub(r"<.*?>", "", text) to clean any web-scraped artefacts.
- Whitespace Normalisation: Multiple consecutive spaces and tabs were collapsed to single spaces and leading/trailing whitespace was trimmed.

3.3 Train / Validation / Test Split

The cleaned dataset was split into three stratified subsets to ensure each partition maintains the same class distribution as the full dataset:

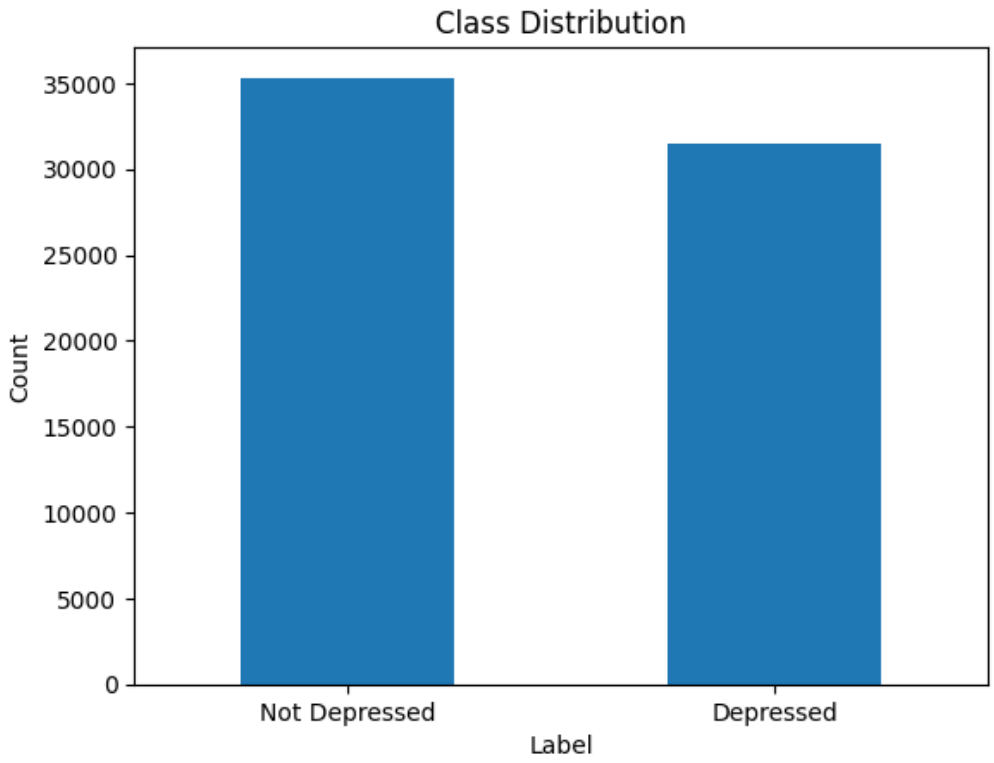
Split	Size	Proportion	Purpose
Training Set	46,733	70%	Model fine-tuning
Validation Set	10,014	15%	Hyperparameter tuning & early stopping
Test Set	10,015	15%	Final unbiased evaluation

4. Exploratory Data Analysis (EDA)

A comprehensive EDA was conducted to understand the dataset's structure, uncover linguistic patterns, and validate the suitability of the data for binary classification. The analysis covers class distribution, text length and word count, frequent word patterns, and special character usage.

4.1 Class Distribution

Plot: Bar Chart Class Distribution

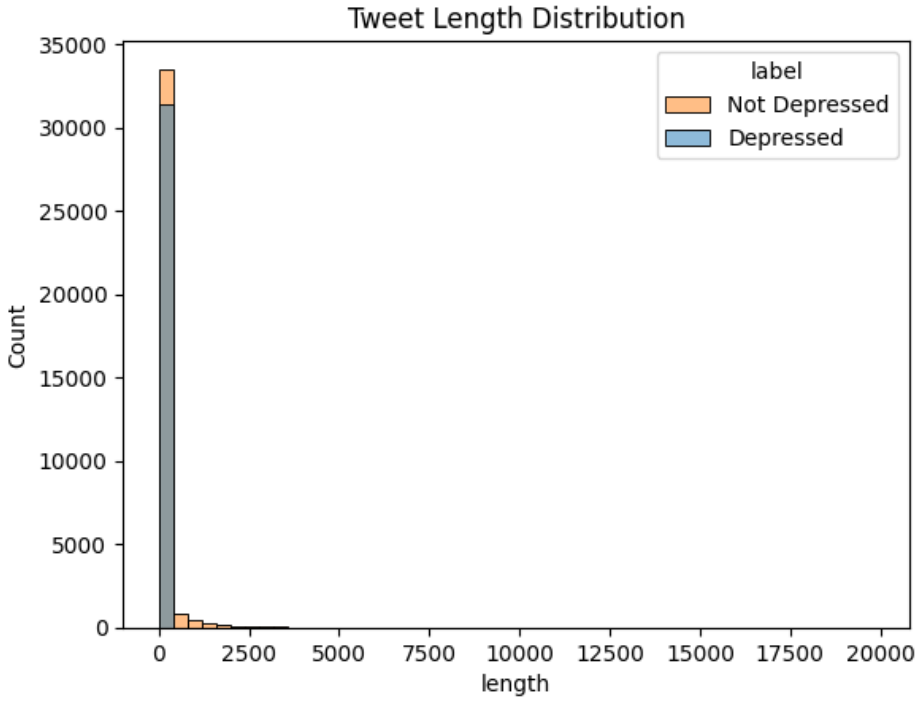


Insight:

- The dataset is well-balanced, with approximately 35,000 samples labelled as Not Depressed and 32,000 labelled as Depressed.
- The near-even class split (~52% vs 48%) is ideal for binary classification as it prevents the model from biasing towards a majority class.
- Because the classes are balanced, Accuracy is a reliable primary metric. In an imbalanced setting, F1-score or AUC would be preferred.

4.2 Text Length Distribution

Plot: Histogram Tweet / Post Length Distribution by Class

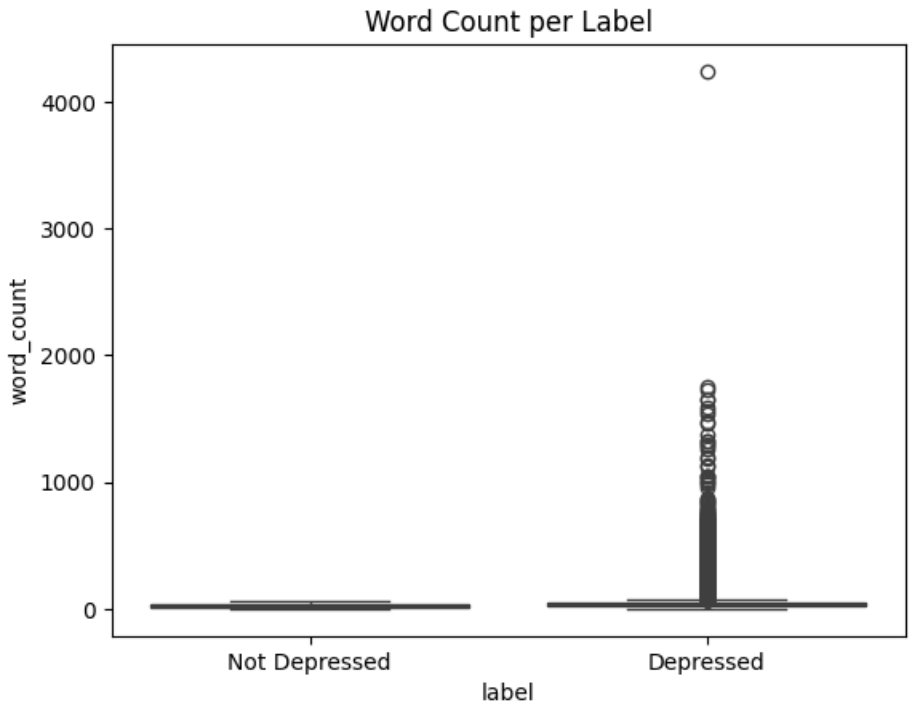


Insight:

- Both classes are heavily right-skewed, meaning the vast majority of posts are under 500 characters typical of social media data.
- The Depressed class shows a notable long tail, with some posts exceeding 10,000–20,000 characters, representing long-form venting or detailed personal narratives.
- Since DistilBERT has a 128-token maximum, long posts are truncated. The first 128 tokens are generally sufficient to capture the emotional tone.

4.3 Word Count per Label

Plot: Box Plot — Word Count per Labe

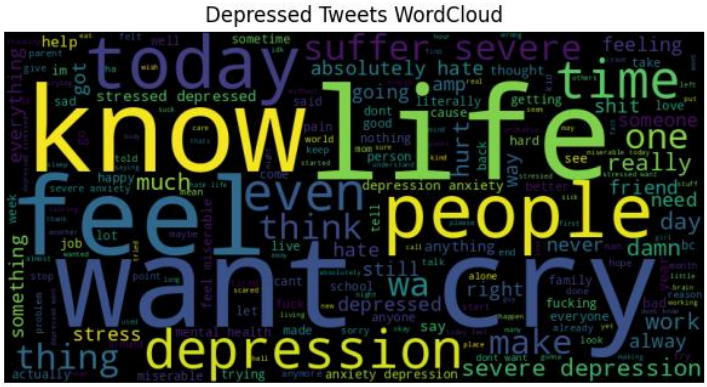


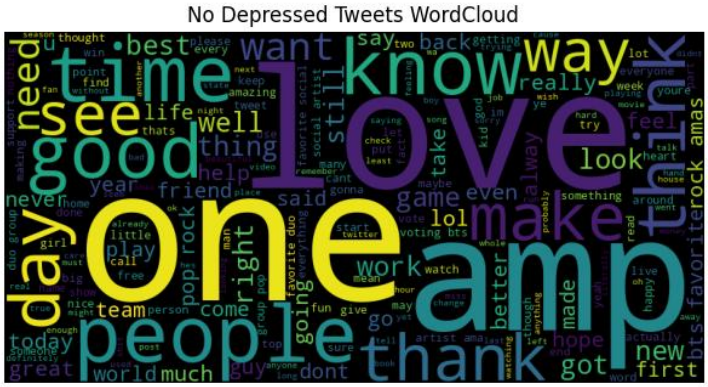
Insight:

- The median word count is low and similar for both classes, indicating that the average post in both categories is concise.
- The Depressed class shows a dramatically wider distribution of outliers, with some posts exceeding 4,000 words a pattern absent in the Not Depressed class.
- This confirms the 'venting phenomenon': individuals expressing depressive symptoms disproportionately use social media for extended personal storytelling.

4.4 Word Frequency Analysis & Word Clouds

Plot: Word Clouds Depressed vs. Not Depressed Classe





Insight:

- Depressed class: dominated by high-intensity emotional tokens such as depression, cry, suffer, miserable, and severe reflecting strong internal distress and unmet emotional needs.
- Not Depressed class: characterised by positive, outward-looking vocabulary including love, good, thank, favorite, and best, indicating social engagement and positive affect.
- Shared words such as time and people appear in both clouds but carry entirely different contextual meanings 'struggling all the time' versus 'having a good time.'

5. Modeling

The modeling phase involved converting the cleaned dataset into HuggingFace Dataset objects, tokenising the text using the DistilBERT tokenizer, and fine-tuning the pre-trained model for binary sequence classification.

5.1 Model Architecture

DistilBERT (distilbert-base-uncased) was selected as the backbone model. It is a distilled version of BERT that retains 97% of its language understanding capability while being 40% smaller and 60% faster. It was adapted for binary sequence classification by appending a linear classification head.

5.2 Tokenisation

Texts were tokenised using the AutoTokenizer for distilbert-base-uncased with the following settings:

- Truncation: True long texts are truncated to the maximum length.
- Padding: max_length all sequences are padded to a uniform length of 128 tokens.
- Max Length: 128 tokens sufficient for the vast majority of social media posts.

5.3 Training Results per Epoch

The best model checkpoint was saved at Epoch 4 (highest F1 = 99.53%).

Epoch	Train Loss	Val Loss	Accuracy	F1	Precision	Recall
1	0.0852	0.0529	99.40%	99.43%	99.64%	99.23%
2	0.0422	0.0596	99.38%	99.41%	99.49%	99.34%
3	0.0173	0.0579	99.41%	99.44%	99.42%	99.47%
4	0.0035	0.0683	99.50%	99.53%	99.58%	99.47%
5	0.0010	0.0794	99.48%	99.51%	99.55%	99.47%

6. Evaluation Results

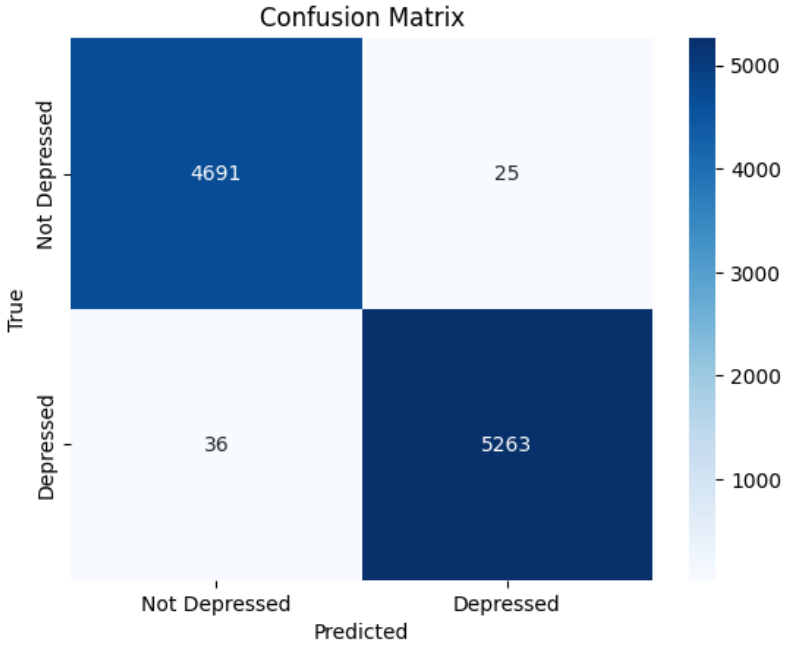
6.1 Internal Test Set Performance

The saved best model was evaluated on the 10,015 held-out test samples that were never seen during training or validation:

Metric	Not Depressed	Depressed	Overall
Precision	0.99	1.00	0.99
Recall	0.99	0.99	0.99
F1-Score	0.99	0.99	0.99
Support	4,716	5,299	10,015
Accuracy	—	—	99.39%

6.2 Confusion Matrix Internal Test Set

Plot: Confusion Matrix Internal Test Set



The confusion matrix shows only 25 false positives and 36 false negatives out of 10,015 predictions, demonstrating exceptional discrimination between the two classes.

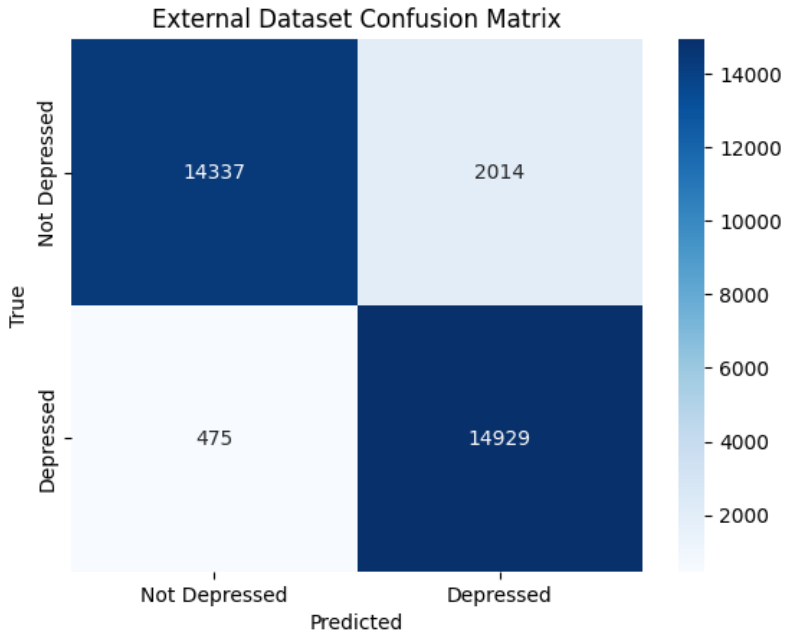
6.3 External Dataset Validation (Kaggle link)

To assess real-world generalisation, the model was evaluated on an independent external dataset (Suchintika Sarkar / Sentiment Analysis for Mental Health), which was not part of the original training pipeline:

Metric	Not Depressed (Normal)	Depressed (Depression)	Overall
Precision	0.97	0.88	0.92
Recall	0.88	0.97	0.92
F1-Score	0.92	0.92	0.92
Support	16,351	15,404	31,755
Accuracy	—	—	92.16%

6.4 Confusion Matrix External Dataset

Plot: Confusion Matrix External Dataset



Insight:

To verify the model's generalizability, it was evaluated on an external dataset. While accuracy dropped from 99% to 92.1%, the model maintained a high Recall of 0.97 for the Depression class. This indicates that the model is highly effective at identifying depressive signals across different data sources, though it tends toward 'over-sensitivity' (False Positives) when analyzing normal everyday stress.

8. System Workflow

MindGuard Pro follows a structured, multi-stage pipeline that takes a user's free-text responses through clinical screening, AI-powered risk classification, and empathetic LLM-generated reporting. The workflow below describes each stage end-to-end, as implemented across the microservices architecture.

8.1 End-to-End Pipeline Overview

The system is composed of seven sequential stages:

1. **User Input Collection** — The Streamlit frontend presents three open-ended domain questions to the user covering Routine & Energy, Social Connection, and Temporal Outlook. The user types free-text responses directly into the interface.
2. **Input Validation & Forwarding** — The frontend validates that all three domain responses are non-empty, then forwards the collected text to the NLP Prediction API (Port 8002) via an asynchronous HTTP POST request.

3. **Text Preprocessing** — The NLP microservice applies the same clean_text() pipeline used during training: URL stripping, HTML tag removal, and whitespace normalisation. This ensures inference consistency with the training distribution.
4. **DistilBERT Classification** — Each of the three preprocessed domain answers is independently tokenised (max 128 tokens) and passed through the fine-tuned DistilBERT model. The model outputs a binary risk label (0 = Not Depressed, 1 = Depressed) along with a confidence score per domain.
5. **Diagnostic Aggregation** — The backend applies the core diagnostic logic: if 2 or more domains return a Depressed label with high confidence, the system flags an Elevated Risk of Depression; otherwise it reports an Optimal Baseline.
6. **RAG-Augmented Report Generation** — The aggregated risk label, confidence scores, and original user responses are sent to the LLM & RAG Generation API (Port 8001). Llama 3.2 retrieves semantically relevant passages from the ChromaDB vector store (seeded from mental health counseling datasets) and generates a personalised clinical insight report grounded in real counseling context.
7. **Frontend Display & Chatbot Session** — The Streamlit frontend renders the diagnostic result, the AI-generated report, and activates the 24/7 AI companion chatbot. The chatbot maintains a stateful session via st.session_state, allowing the user to continue a context-aware conversation about their results.

8.2 Pipeline Architecture Diagram

The diagram below summarises data flow across system components:

```
[ Streamlit Frontend (app.py) ]
| HTTP POST |
v           v
[ NLP API :8002 ]   [ RAG API :8001 ]
DistilBERT Model    Llama 3.2 + ChromaDB
|                   |
Risk Labels         Clinical Report + Chatbot
|_____________________________________|
|
[ Dashboard + AI Chatbot ]
```

8.3 Microservices Communication

The two backend services operate independently and can be scaled or replaced without affecting each other. The table below summarises their responsibilities:

Service	Port	Responsibility
NLP Prediction API	8002	Preprocesses text, runs DistilBERT inference, returns binary risk labels per domain
LLM & RAG API	8001	Hosts Llama 3.2, queries ChromaDB vector store, generates personalised clinical insight reports
Streamlit Frontend	8501 (default)	Collects input, orchestrates API calls, renders dashboard, manages chatbot session state

9. System Design and Interface

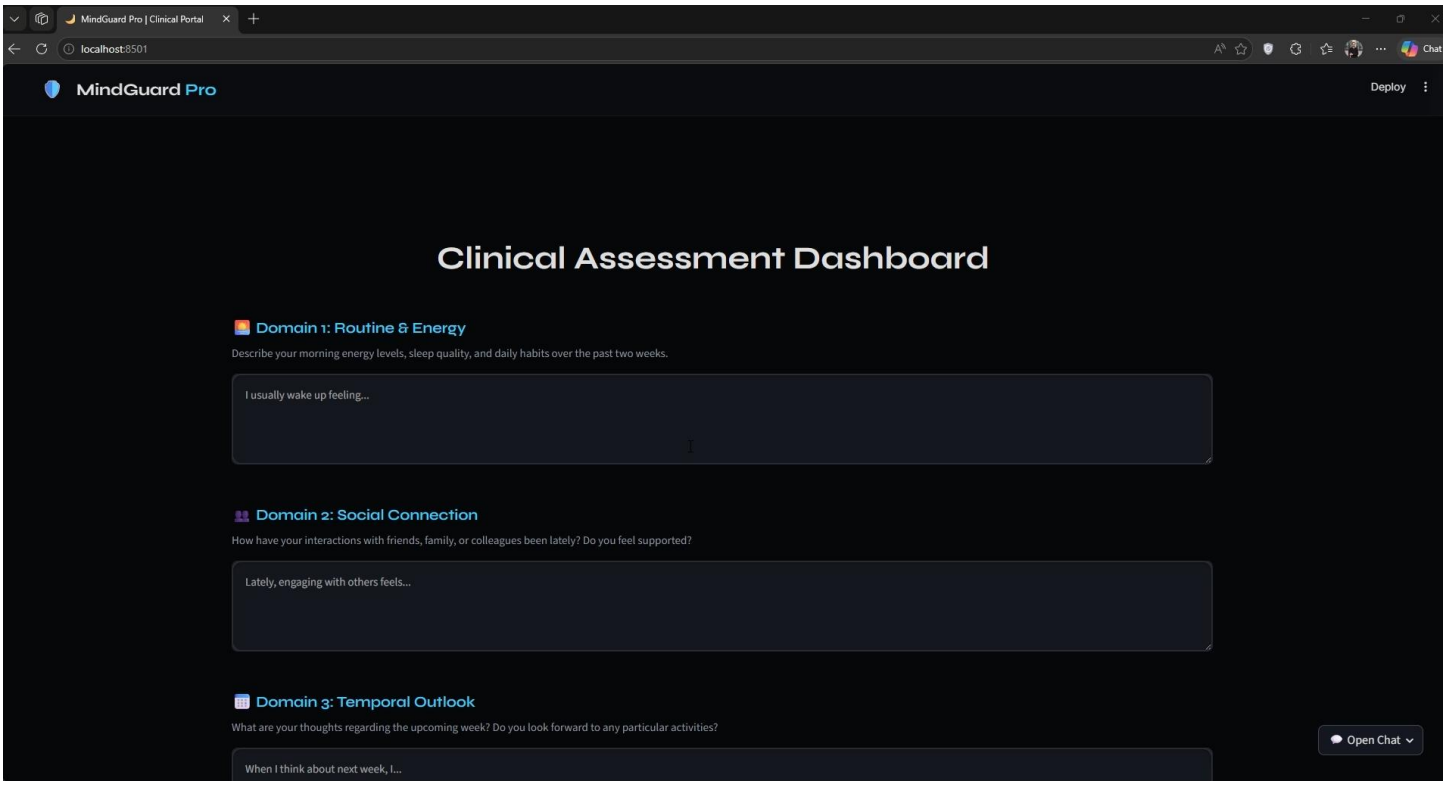
MindGuard Pro's frontend is built with Streamlit and enhanced with enterprise-grade custom CSS to deliver a clinical, professional aesthetic. This section describes the key interface components and the design decisions behind each.

9.2 Main Dashboard (Diagnostic Screening)

The primary interface presents the three-domain clinical questionnaire. Each domain is rendered as a distinct card with a descriptive label and a free-text area:

- **Domain 1 — Routine & Energy:** Captures morning energy levels, sleep quality, and daily habits.
- **Domain 2 — Social Connection:** Assesses interactions with friends and family, and feelings of isolation or support.
- **Domain 3 — Temporal Outlook:** Explores future-oriented thinking, anticipation, and emotional dread regarding the upcoming week.

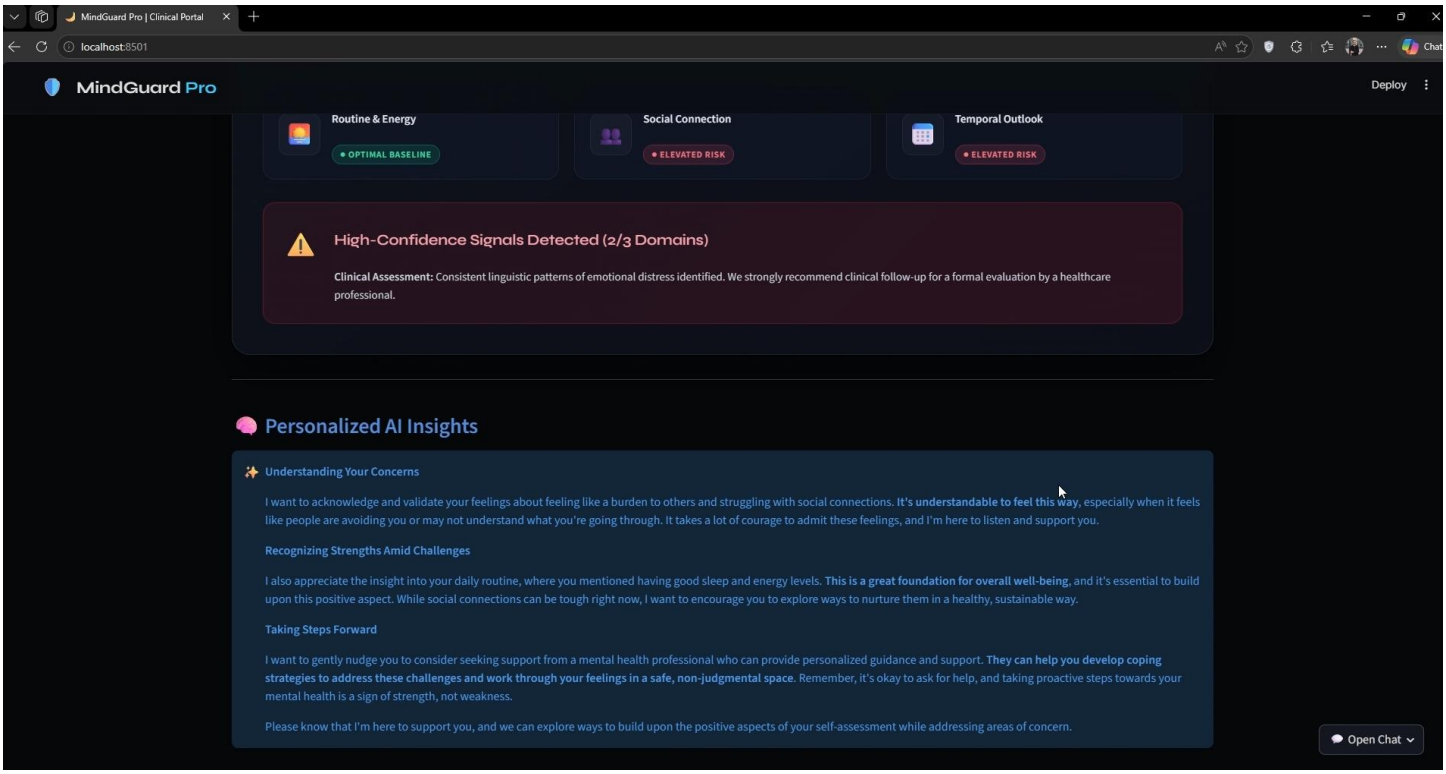
After submission, the dashboard transitions to display the diagnostic result prominently either an Elevated Risk of Depression alert (styled in red) or an Optimal Baseline confirmation (styled in green) followed by the AI-generated clinical insight report.



9.3 Diagnostic Result Display

The result panel dynamically renders based on the aggregation logic outcome:

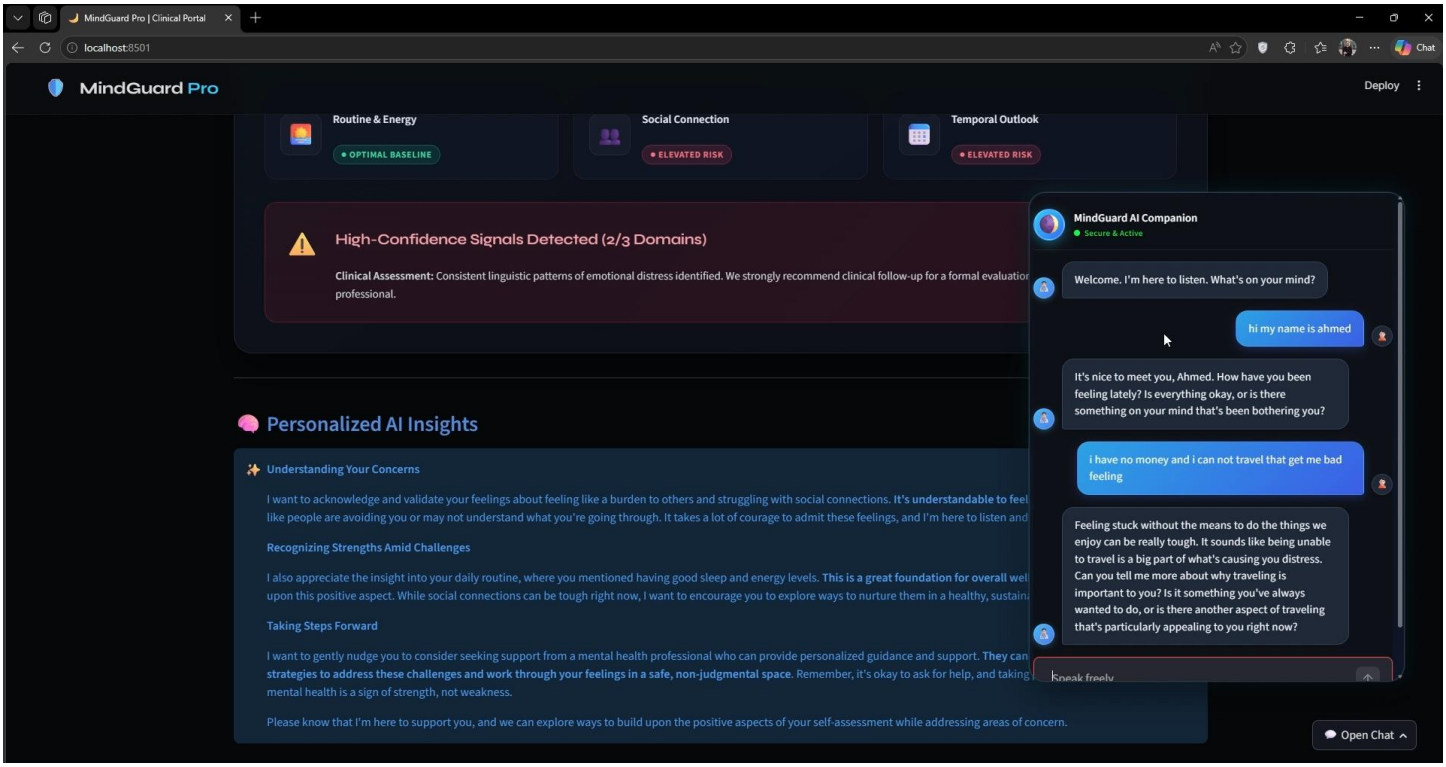
- **Elevated Risk Card:** Displays a prominent red-accented warning banner listing the flagged domains, the confidence scores per domain, and a disclaimer encouraging professional consultation.
- **Optimal Baseline Card:** Displays a green-accented confirmation with per-domain confidence scores and a positive reinforcement message.
- **AI Clinical Report Section:** Below the result card, Llama 3.2's RAG-grounded narrative report is streamed token-by-token, giving the interface a live 'typing' effect that improves perceived responsiveness.



9.4 AI Companion Chatbot

The chatbot panel is the centerpiece of the MindGuard Pro experience, providing 24/7 empathetic, context-aware emotional support:

- **RAG-Grounded Responses:** The Llama 3.2 model retrieves relevant passages from the ChromaDB vector store populated with real mental health counseling conversations before generating each reply, ensuring responses are grounded in evidence-based counseling language rather than generic LLM output.
- **Context Continuity:** The chatbot is initialised with the user's diagnostic result and original domain responses as hidden context, allowing it to reference the user's specific situation without the user needing to re-explain.
- **Conversation History:** All messages are persisted in st.session_state and displayed in a scrollable chat window styled to differentiate user messages (right-aligned, blue) from assistant responses (left-aligned, dark card).
- **Safe Messaging Guidelines:** The system prompt explicitly instructs Llama 3.2 to follow safe messaging guidelines: never diagnose, always encourage professional help, and respond with empathy rather than clinical authority.



10. Conclusion and Possible Improvements

10.1 Conclusion

MindGuard Pro demonstrates that high-accuracy, clinically meaningful depression screening is achievable through fine-tuned transformer models trained on diverse social media datasets. The fine-tuned DistilBERT model achieved 99.39% accuracy on the internal test set and 92.16% on an entirely independent external dataset maintaining a critical Depression-class Recall of 0.97, meaning that genuine distress signals are rarely missed.

Beyond classification accuracy, the project successfully integrates a multi-component AI stack into a coherent, user-facing product. The microservices architecture decouples the NLP inference layer from the LLM generation layer, making each independently scalable and maintainable. The RAG pipeline grounds the chatbot's responses in real counseling context, elevating the quality of support beyond what a generic LLM prompt could achieve.

The choice of DistilBERT retaining 97% of BERT's language understanding at 60% of the inference cost reflects a deliberate balance between clinical accuracy and deployment practicality. The system is designed to serve as an intelligent early-screening tool, not a diagnostic replacement for licensed mental health professionals, and its interface enforces this distinction throughout.

10.2 Limitations

Despite strong performance, the following limitations should be acknowledged:

- **128-Token Truncation:** Long-form posts (common in the Depressed class) are truncated to the first 128 tokens. Critical emotional signal appearing mid-post or near the end may be lost during inference.
- **Free-Text Domain Framing:** The three domain questions are framed for non-clinical users. However, a user deliberately masking their symptoms could produce misleading results, as the model cannot detect intentional concealment.
- **Local LLM Dependency:** Llama 3.2 is hosted locally via Ollama, requiring significant compute resources (GPU recommended). This limits accessibility for users without appropriate hardware.

10.3 Possible Improvements

The following enhancements are proposed as directions for future development:

- **Longer Context Window:** Replace the 128-token limit with a longer-context model (e.g., Longformer or Clinical-BERT) to fully process long-form venting posts without truncation.
- **Multi-Label Classification:** Extend beyond binary depression detection to multi-label classification covering co-occurring conditions such as anxiety, PTSD, and bipolar disorder.
- **Cross-Lingual Support:** Fine-tune on multilingual datasets (using XLM-RoBERTa or mBERT) to extend screening capability to Arabic, French, and other languages relevant to the target region.

Summary

MindGuard Pro establishes a strong technical foundation for AI-assisted mental health screening. With targeted improvements in model robustness, clinical alignment, and deployment infrastructure, it has the potential to evolve into a production-grade clinical decision-support tool that meaningfully complements rather than replaces professional mental healthcare.